Rakamin Academy

Bank Muamalat

Sales Data Analytics and Interactive Dashboard Using BigQuery and Looker Studio

PT Sejahtera Bersama

Project-Based Virtual Intern: Business Intelligence Analyst x Rakamin Academy

Presented by
M Hasbi Asshiddiqqi

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Hello I am,
M Hasbi Asshiddiqqi

Information Systems graduate from Telkom University (GPA 3.69) with hands-on experience in network device monitoring and data management during an internship at PT Telkom Indonesia. Skilled in troubleshooting hardware and software, providing end-user support, and maintaining IT inventory. Proficient in monitoring network infrastructure (OLT, ODC, ODP) and using Microsoft Excel for data compilation and reporting. Detail-oriented, proactive, and communicative, with a strong commitment to ensuring efficient IT operations and delivering reliable user support in a professional environment.

Professional Experience

PT Aplikasi Kriya Utama (Appskep Indonesia)

***Creative Officer Intern* | Sep 2025 - Jan 2026**

- Produced 70+ visual contents using Figma and Affinity Designer, achieving 1,000,000+ total views within 5 months
- Conducted A/B testing various content types and design styles to identify most effective visual elements for increasing reach and engagement.

PT Telkom Indonesia (Witel Sumbar)

***AODM Intern* | Jun 2024 - Aug 2024**

- Monitored and verified maintenance of approximately 50 network devices weekly (OLT, FTM, ODC, ODP) performed by field technicians, ensuring compliance with operational standards.
- Compiled UNSC data using Microsoft Excel (VLOOKUP) with volume of ±250 rows per month, consistently completing daily targets in under 2 hours.

CASE STUDY

During the Project-based Business Analyst Intern program at Bank Muamalat, I was trained to become a proficient Business Intelligence Analyst through various learning materials and assignments. However, theoretical knowledge alone is insufficient; becoming a true industry professional requires hands-on practice and a tangible portfolio as proof of competency. To address this gap, a final project was assigned involving raw datasets to be transformed into meaningful data visualizations. Bank Muamalat provided a case study of PT Sejahtera Bersama, a company struggling to maintain and increase its sales. The company needs to understand its sales performance across different product categories and geographic regions to identify areas for improvement and growth. Specifically, a comprehensive sales analysis supported by data visualization is required to uncover patterns, highlight top-performing and underperforming segments, and provide actionable insights. Therefore, this project aims to develop an interactive sales dashboard complete with business insights and strategic recommendations, serving as a decision-making tool for the company to optimize its sales strategy and ultimately boost future revenue.

How can raw transaction data be visualized into an informative dashboard?

What business insights and strategic recommendations to retain customers and increase sales?

How can this project serve as a portfolio demonstrating readiness to work in the industry as a Business Intelligence Analyst?

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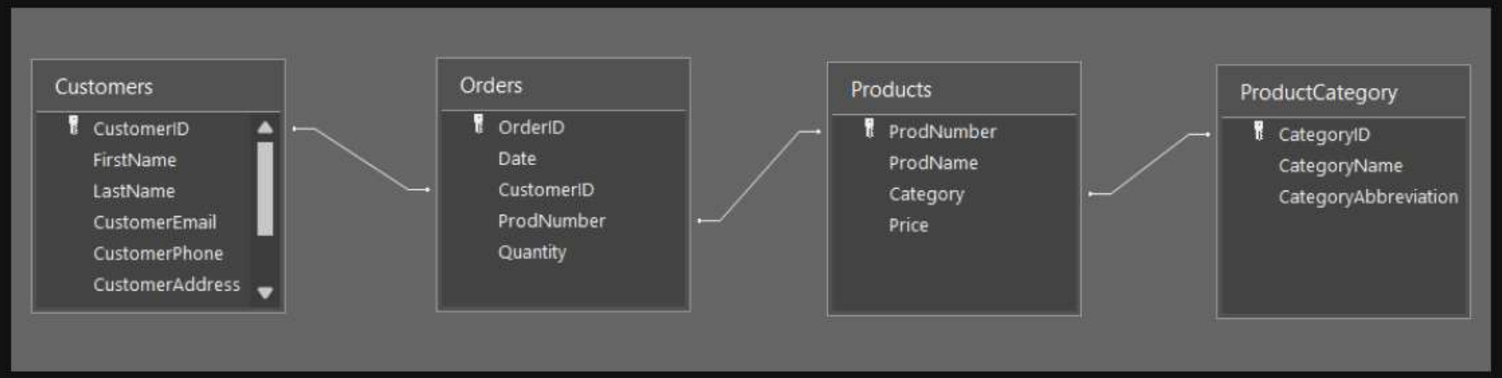
Data Key Definition > Data Relationship Design > Data Integration > Data Visualization

Table	Key Columns
Customers	**CustomerID (PK) (INT)**
	FirstName (VARCHAR)
	LastName (VARCHAR)
	CustomerEmail (VARCHAR) (UNIQUE)
	CustomerPhone (VARCHAR)
	CustomerCity (VARCHAR)
	CustomerState (VARCHAR)
	CustomerZip (Varchar)
Orders	**OrderID (PK) (INT)**
	Date (DATE)
	CustomerID (FK, NOT NULL) (INT)
	ProdNumber (FK, NOT NULL) (VARCHAR)
	Quantity (CHECK) (INT)
Products	**ProdNumber (PK) (VARCHAR)**
	ProdName (VARCHAR)
	Category (FK) (INT)
	Price (DECIMAL)
ProductCategory	**CategoryID (PK) (INT)**
	CategoryName (VARCHAR)
	CategoryAbbreviation (VARCHAR)

Four main datasets serve as the foundation for analyzing business problems: **Customers, Orders, Products,** and **ProductCategory**. First, the Customers dataset stores personal information about customers. Second, the Orders dataset records information related to purchase transactions. Third, the Products dataset captures everything related to the product catalog. Finally, the ProductCategory dataset plays a role in grouping products into more structured categories.



These datasets are interconnected, creating a seamless flow of business process information. Customers is linked to Orders through **'CustomerID'** as the Primary Key, which uniquely identifies customers in every order. Next, Orders is connected to Products via **'ProdNumber'** as the Primary Key, uniquely identifying each item in a purchase. Lastly, Products is linked to ProductCategory through **'Category/ CategoryID'** as the Primary Key, ensuring each product is grouped according to its type to support business operations efficiently.

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Data Integration & ETL Using Google BigQuery

Products | *BI Analyst

BI Analyst | Run | Download | Share | Schedule | Open in | More | Save query

```sql
SELECT
  o.Date as order_date,
  pc.CategoryName as category_name,
  p.ProdName as product_name,
  p.Price as product_price,
  o.Quantity as order_qty,
  (p.Price * o.Quantity) as total_sales,
  c.CustomerEmail as cust_email,
  c.CustomerCity as cust_city

FROM `rakaminmuamalat.ptsejahteraanalysis.Orders` AS o
LEFT JOIN `rakaminmuamalat.ptsejahteraanalysis.Products` AS p ON p.ProdNumber = o.ProdNumber
LEFT JOIN `rakaminmuamalat.ptsejahteraanalysis.ProductCategory` AS pc ON pc.CategoryID = p.Category
LEFT JOIN `rakaminmuamalat.ptsejahteraanalysis.Customers` AS c ON c.CustomerID = o.CustomerID
```

Query completed

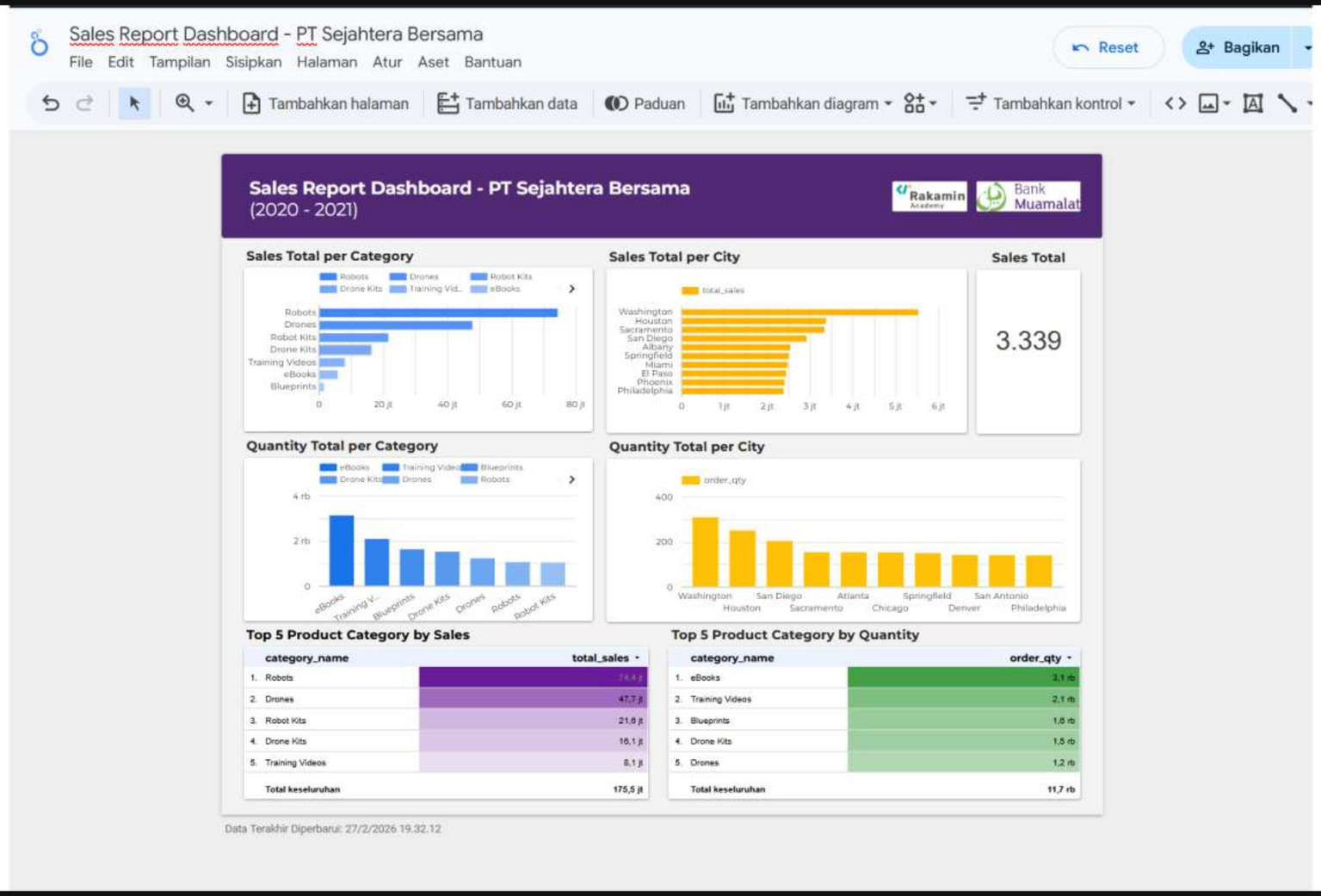
Query results | + Create conversation | Save results | Open in

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	order_date	category_name	product_name	product_price	order_qty	total_sales	cust_email	cust_city
1	2020-01-01	eBooks	Polar Robots	2399	2	4798	fvaslerqt@comsenz.com#mailt...	Jackson
2	2020-01-01	Robots	RWW-75 Robot	88300	3	264900	tmckernot@tinyurl.com#mailto:...	Katy
3	2020-01-01	Training Videos	Drone Video Techniques	3799	6	22794	gstiggersdd@eventbrite.com# mailto:gstiggersdd@eventbrite	Saint Petersburg

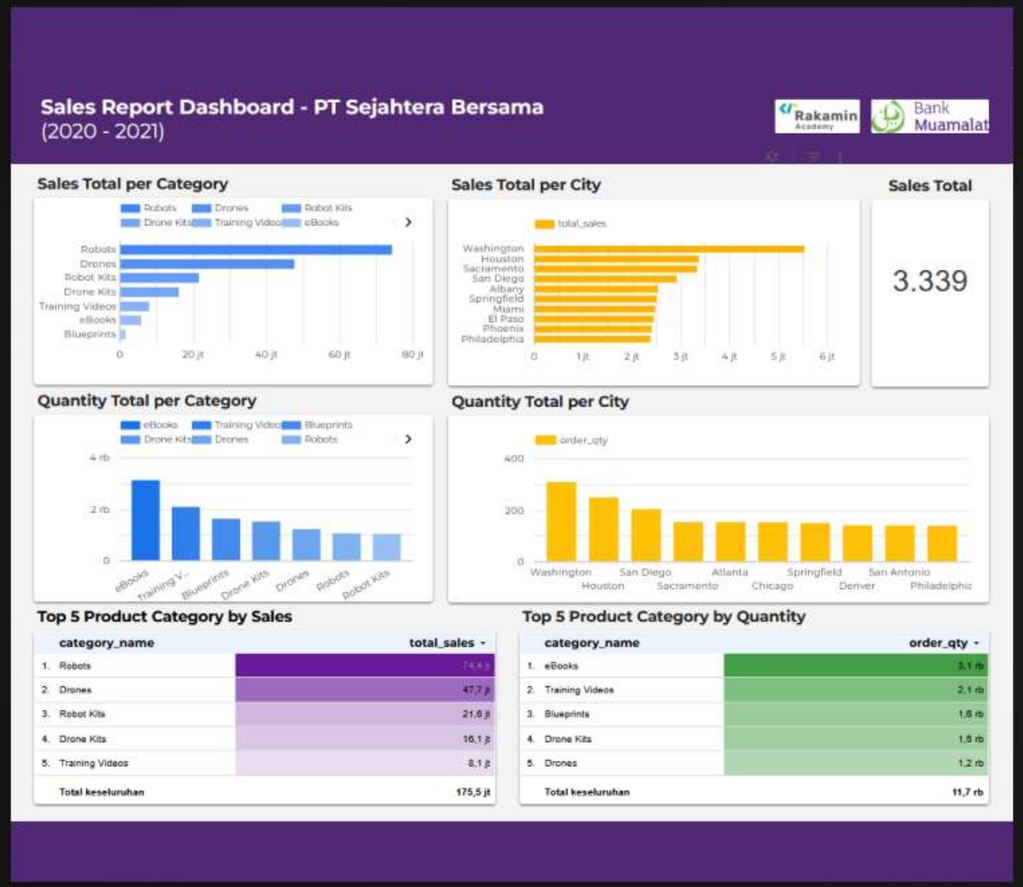
Results per page: 50 | 1 – 50 of 3339

Data Visualization Using Looker Studio



RESULTS

Problem #1: How can raw transaction data be visualized into an informative dashboard?



A "Sales Report Dashboard" was developed to transform the available aggregate data into intuitive visualizations. This dashboard provides a comprehensive overview of PT Sejahtera Bersama sales performance accross product categories and geographic regions, enbaling management to monitor key metrics and identify patterns at a glance.

Key Insight

- Sales reached 3.339 in total serving as the primary metric.
- Robots and Drones are top performing Category in Total of Sales.
- eBooks and Training Video are top performing Category in Total of Quantity.
- The top performing cities by Sales are Washington, Houston and Sacramento.
- The top performing cities by Quantity are Washington, Houston and San Diego.

RESULTS

Problem #2: What business insights and strategic recommendations to retain customers and increase sales?

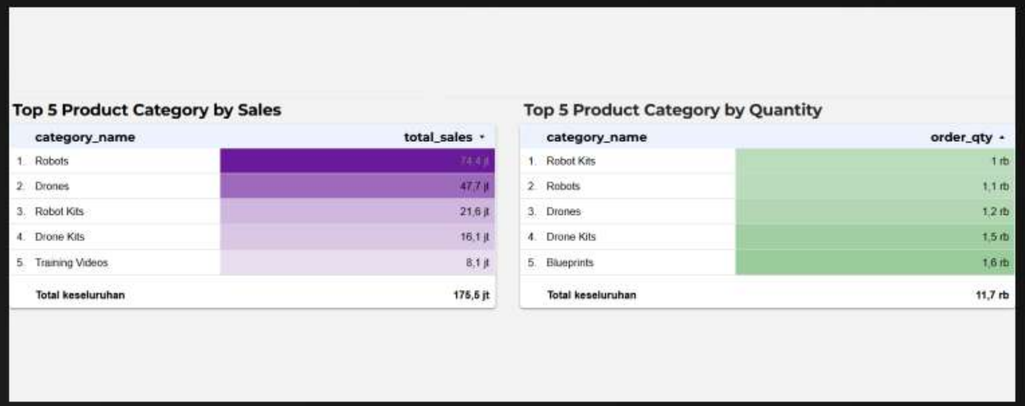
Based on the sales dashboard analysis, there are several key business insights have been identified. These insights reveal important patterns in product performance, customer buying behavior and geographic distribution that can guide strategic decision-making at PT Sejahtera Bersama.

Insight #1: High-Value Products Drive Highest Revenue but Have The Lowest Transaction Frequency

- **Robots** and **Drones** alone contribute nearly **70%** of total revenue from top categories.
- However, these categories also have the lowest transaction volume **(nearly 20%)** of total product category purchasing.

Insight #2: Digital Products Drive The Highest Engagement but Low in Revenue

- **eBooks** and **Training Videos** have the highest transaction volume **(3.1k and 2.1k respectively)** but generate minimal revenue comparte to any physical products..

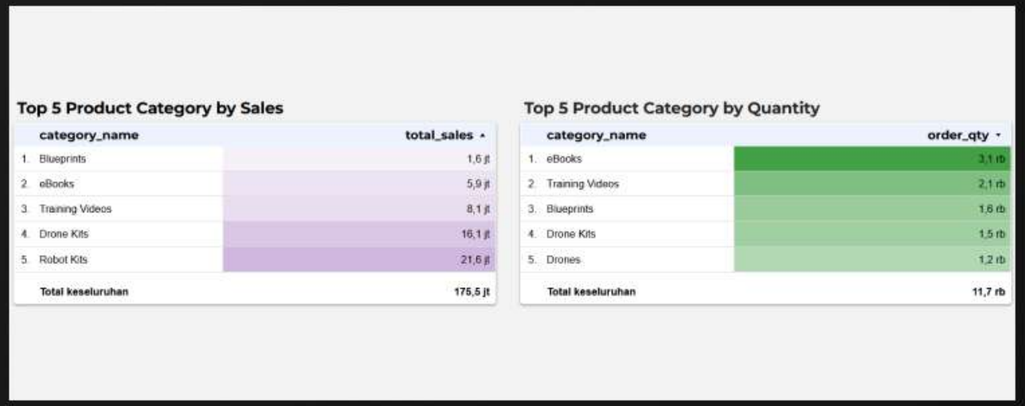


Top 5 Product Category by Sales

	category_name	total_sales
1.	Robots	74,4 jt
2.	Drones	47,7 jt
3.	Robot Kits	21,6 jt
4.	Drone Kits	16,1 jt
5.	Training Videos	8,1 jt
	Total keseluruhan	175,5 jt

Top 5 Product Category by Quantity

	category_name	order_qty
1.	Robot Kits	1 rb
2.	Robots	1,1 rb
3.	Drones	1,2 rb
4.	Drone Kits	1,5 rb
5.	Blueprints	1,6 rb
	Total keseluruhan	11,7 rb



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RESULTS

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Insight #3: Two Possible Distinct Customer Segments Exist

- **Segment A (High-Value Buyers)** purchase physical product who spend large amounts of money but transact infrequently.
- **Segment B (High-Frequency Buyers)** purchase digital product who transact frequently but spend small amounts of money.

Insight #4: Sales are Geographically Concentrated

- **Washington, Houston** and **Sacramento** are the top 3 cities for the majority of sales.
- **Philadelphia** shows are the lowest both on sales revenue and total transaction.

Insight #5: A Significant Gap Between Sales and Quantity in Some Categories

- **Robots** and **Drones** generate highest sales but are not in the top 5 quantity categories.
- **eBooks** and **Training Videos** generate highest quantity but are not in top 5 sales categories.

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Problem #2: What business insights and strategic recommendations to retain customers and increase sales?

Based on the key business insights that has been identified from the sales dashboard, the following recommendations are proposes for PT Sejahtera Bersama. These recommendations are designed to address current challenges, capitalize on opportunities and drive sustainable revenue growth. The recommendations are categorized and divided into 3 categories, that are short-term, mid-term, and long-term solutions.

Short-Term Recommendations

- Implement **cross-selling products bundling** between physical and digital products.
- Run some ads or campaigns on specific city that has lowest both sales and quantity transactions and offer free shipping or special discounts.

Mid-Term Recommendations

- Create and develop themed products bundles like (robots + drones + robot kit) and offer bundle special pricing compared to buying items individually.
- Conduct market surveys for the top 5 underperforming cities to understand local preferences and barriers.

Long-Term Recommendations

- Develop a **Customer Segmentation Strategy** by build a customer database based on purchase history into several categories.
- Expand digital product offerings with high engagement but low in revenue by giving premium digital offerings.

RESULTS

Problem #3: How can this project serve as a portfolio demonstrating readiness to work in the industry as a Business Intelligence Analyst?

This project serves as a comprehensive demonstration of the skills and competencies required for a Business Intelligence Analyst role in the industry. Beyond delivering actionable insights for PT Sejahtera Bersama, this work represents a tangible portfolio piece that showcases the ability to transform raw data into strategic business value.

Proof of Competency

- Create a professional industry standard data visualizations.
- Generate and develop business insights along with strategic recommendations.

Industry Readiness

- End-to-end project execution from preparing raw data to visualization with business problem solution.
- Understanding of business metrics and performance indicators and capable to think strategically about business problems.

Unique Selling Points from Others

- Hands-on experience with a semi-realistic business case study.
- Ability to deliver value with very limited data.
- Structured approach to problem-solving business case.
- Balance of technical and aesthetic of creative business presentation.

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Thank You

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